

Electromagnetic Spectrum & Applications

A Complete Lesson Guide

Grade 6 • STEM • Physics

Subject	STEM / Physics
Grade	Class 6
Topic	Electromagnetic Spectrum & Applications
Duration	60–75 Minutes
Focus	Radio, Microwave, IR, Visible, UV, X-Ray, Gamma

*Fun Fact: EM waves travel at 300,000 km/s — the speed of light!
Radio waves can be longer than a football field; gamma rays are smaller than an atom's nucleus*

1. Learning Objectives

Define	the electromagnetic spectrum and explain its components.
Identify	the 7 types of EM waves in order of wavelength and frequency.
Explain	why each type of wave is used for specific real-world applications.
Compare	wavelength, frequency and energy across the spectrum.
Evaluate	safety considerations for different types of EM radiation.

2. What Is the Electromagnetic Spectrum?

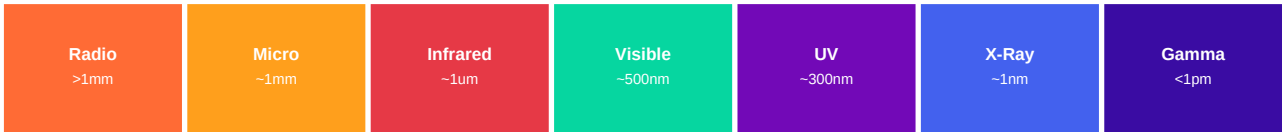
The **electromagnetic spectrum** is the complete range of all electromagnetic radiation — all the different types of light energy that travel as waves through space. From the longest radio waves to the shortest gamma rays, they are all the same type of wave, just with different wavelengths and frequencies.

EM radiation consists of oscillating electric and magnetic fields travelling together. They **do not need a medium** to travel — they can pass through the vacuum of space at the speed of light (3×10^8 m/s). The key difference between each type is their **wavelength** (distance between wave peaks) and **frequency** (waves per second, measured in Hertz).

Wavelength (λ) Distance between two consecutive wave peaks. Measured in metres (m), nanometres (nm) or picometres (pm).	Frequency (f) Number of waves passing a point per second. Measured in Hertz (Hz). Higher frequency = shorter wavelength.
Energy (E) $E = hf$ (Planck's equation). Higher frequency = more energy. Gamma rays carry the most; radio waves the least.	Speed of Light (c) $c = f \times \lambda$. All EM waves travel at 3×10^8 m/s in a vacuum. Speed stays constant; only f and λ change.

3. The Complete EM Spectrum

The spectrum is arranged by wavelength — from longest (radio) to shortest (gamma). As wavelength decreases, frequency and energy increase. Visible light is just a tiny sliver of the entire spectrum!



Longer wavelength / Lower frequency / Less energy

Higher energy / Higher frequency / Shorter wavelength

Wave Type	Wavelength	Frequency	Energy	Key Uses
Radio	>1 mm	< 3 GHz	Lowest	Broadcasting, WiFi, Mobile phones
Microwave	1mm - 10cm	3GHz - 300GHz	Low	Cooking, Radar, Satellite comms
Infrared	700nm - 1mm	300GHz-430THz	Medium	TV remotes, Thermal cameras, Heat
Visible	380-700 nm	430-770 THz	Medium	Human vision, Photography, Fibre optics
Ultraviolet	10-380 nm	770THz-30PHz	High	Sterilisation, Sun tanning, Counterfeit detection
X-Ray	0.01-10 nm	30PHz-30EHZ	Very High	Medical imaging, Airport security, NDT
Gamma Ray	< 0.01 nm	> 30 EHz	Highest	Cancer treatment, Sterilisation, Astronomy

4. Each Wave Type In Depth

Radio

1

Radio Waves

Wavelength: > 1 millimetre**Key Applications:**

- Radio & TV broadcasting
- Mobile phone signals
- WiFi & Bluetooth

Safety: Safe for everyday use — used constantly around us.

Microw

2

Microwaves

Wavelength: 1 mm – 10 cm**Key Applications:**

- Microwave ovens (heat food)
- Weather radar systems
- Satellite communication

Safety: Safe when properly shielded; ovens have metal mesh screens.

Infrar

3

Infrared (IR)

Wavelength: 700 nm – 1 mm**Key Applications:**

- TV & appliance remote controls
- Night-vision cameras
- Heat sensors & thermal imaging

Safety: Near-IR is safe; intense IR (heat) can cause burns.

Visibl

4

Visible Light

Wavelength: 380 nm – 700 nm**Key Applications:**

- Human vision & photography
- Optical fibre internet
- Laser surgery

Safety: Generally safe; intense laser beams can damage eyes.

Ultrav

5

Ultraviolet

Wavelength: 10 nm – 380 nm**Key Applications:**

- Sterilising medical equipment
- Sun tanning / sunburn protection
- Detecting counterfeit banknotes

Safety: Wear SPF sunscreen outdoors; overexposure causes skin cancer.

X-Rays

6

X-Rays

Wavelength: 0.01 nm – 10 nm**Key Applications:**

- Medical bone & tissue imaging
- Airport security scanners
- Non-destructive testing of metals

Safety: Dangerous in high doses — doctors use lead shields and limit exposure.

Gamma

7

Gamma Rays

Wavelength: $< 0.01 \text{ nm}$

Key Applications:

- Cancer radiotherapy (kills tumour cells)
- Sterilising food & medical tools
- Studying supernovas in space

Safety: Most dangerous — radioactive sources; requires thick lead/concrete shielding.

5. Why Each Wavelength Is Used Where It Is

The *reason* each wave type is used for a specific purpose comes directly from its **physical properties** — wavelength, frequency, energy, and how it interacts with matter.

Wave	Physical Property	Therefore Used For
Radio	Long wavelength passes through walls and atmosphere without absorption	Broadcasting, WiFi — signals reach everywhere without obstacles
Microwave	Specific frequency absorbed by water molecules, causing them to vibrate	Cooking (heats food from inside); radar (bounces off objects)
Infrared	All objects above absolute zero emit IR; warm bodies emit more strongly	Thermal cameras detect heat; remotes send short-range IR pulses
Visible	Wavelength our retinas evolved to detect; colours separated by prism	Vision, photography, fibre optics (total internal reflection)
UV	High enough energy to break chemical bonds and damage DNA	Kills bacteria (sterilisation); detects fluorescent security markings
X-Ray	Short wavelength passes through soft tissue but blocked by denser bone	Medical imaging — bones appear white, soft tissue appears dark
Gamma	Highest energy — penetrates almost everything; kills living cells	Radiotherapy targets and kills tumour cells; sterilises equipment

Visible Light: The Rainbow

Visible light is the only part of the EM spectrum our eyes can detect. White light (like sunlight) is actually a mixture of all colours. When it passes through a prism or water droplet, it separates into a spectrum of colours — the **rainbow**!

Red 700nm	Orange 620nm	Yellow 580nm	Green 530nm	Blue 470nm	Indigo 445nm	Violet 400nm
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ROY G BIV — Red, Orange, Yellow, Green, Blue, Indigo, Violet — ordered by increasing frequency

6. Safety with Electromagnetic Waves

Wave Type	Safety Level	Precaution Needed
Radio Waves	SAFE	None — used constantly for phones, WiFi, radio
Microwaves	SAFE	Metal mesh shields inside microwave ovens
Infrared	GENERALLY SAFE	Avoid staring at intense IR sources (eg. infrared heaters)
Visible	GENERALLY SAFE	Avoid intense laser beams — can damage retina
Ultraviolet	MODERATE RISK	Apply SPF sunscreen; UV-protective glasses outdoors
X-Rays	HIGH RISK	Lead aprons for patients; limited medical use only
Gamma Rays	VERY HIGH RISK	Thick lead or concrete shielding; strictly controlled

7. Amazing Fun Facts

01	Speed of Light EM waves travel at 299,792,458 m/s — so fast that light from the Sun reaches Earth in just 8 minutes 20 seconds!
02	Giant Radio Waves Some radio waves used for long-range communication can be longer than 10 kilometres!
03	Animal Superpowers Bees can see ultraviolet light and use UV patterns on flowers to find nectar. Pit vipers detect infrared to hunt prey in the dark!
04	Space Explosions Gamma rays come from the most energetic events in the universe — supernovae, neutron star collisions, and black holes!
05	Cosmic Background The universe is filled with microwave radiation left over from the Big Bang 13.8 billion years ago — the 'echo' of creation!
06	Visible Slice The visible light spectrum is only about 0.0035% of the entire electromagnetic spectrum. We are essentially colour-blind to the universe!

8. Practice Questions

Answer the following questions to check your understanding of the EM spectrum.

Q1. What is the relationship between wavelength and frequency in EM waves?

- A) They are always equal
- B) As wavelength increases, frequency increases
- C) As wavelength increases, frequency decreases
- D) They are completely unrelated

Answer: C) As wavelength increases, frequency decreases ($c = f \times \lambda$, speed is constant)

Q2. A doctor needs to see a patient's broken bone without surgery. Which type of EM radiation should be used?

- A) Infrared
- B) Radio Waves
- C) Ultraviolet
- D) X-Rays

Answer: D) X-Rays — they pass through soft tissue but are blocked by denser bone material

Q3. Why are microwaves used to heat food rather than radio waves?

- A) Microwaves travel faster than radio waves
- B) Microwaves have a specific frequency absorbed by water molecules
- C) Microwaves are cheaper to produce
- D) Radio waves cannot enter the kitchen

Answer: B) Microwaves are absorbed by water molecules, causing them to vibrate and generate heat

Q4. Which part of the EM spectrum has the HIGHEST energy?

- A) Radio Waves
- B) Visible Light
- C) Ultraviolet
- D) Gamma Rays

Answer: D) Gamma Rays — they have the shortest wavelength and highest frequency

Q5. Why should you apply sunscreen before going outdoors?

- A) To block radio waves
- B) To block infrared heat from the sun
- C) To protect skin from UV radiation which can damage DNA
- D) To block visible light

Answer: C) UV radiation has high enough energy to break chemical bonds in DNA, potentially causing skin cancer



9. Lesson Summary

RADIO	MICRO	INFRARED	VISIBLE	UV	X-RAY	GAMMA
> 1mm	~1mm	~700nm	380-700nm	~300nm	~1nm	< 1pm
Lowest Energy	Low Energy	Medium	Medium	High	V.High	Highest Energy
WiFi Radio TV	Ovens Radar Satellite	Remotes Thermal Night vision	Vision Photography Fibre	Sterilise Sunscreen Fake notes	Medical Security NDT	Cancer Therapy Astronomy

10. Key Takeaways

01

The EM spectrum includes many wave types beyond visible light — radio, microwave, IR, UV, X-ray and gamma.

02

All EM waves travel at the speed of light (3×10^8 m/s) but differ in wavelength and frequency.

03

Shorter wavelength = higher frequency = higher energy. This determines what each wave is used for.

04

Each wave type's physical properties make it ideal for specific applications.

05

Safety increases with energy: radio waves are safe; gamma rays require heavy shielding.

Keep exploring! Look for electromagnetic waves in your daily life — from the WiFi connecting you to the internet, to the microwave heating your food, to the sunscreen protecting your skin from UV rays. EM waves are everywhere!